

Computer Graphics I

4003-570 / 4005-761

Midterm Project Option

Date posted: March 29th 2013

Due date: April 12th 2013

Submit to the appropriate dropbox on myCourses.

Purpose:

In this project, you will implement the complete 2D pipeline as presented in class using routines written for previous assignments.

Goals:

- Test your knowledge of the components and algorithms of 2D drawing
- Combine existing code into a complete pipeline application

2D Programming Environment:

Like the assignments, the programming environment that you will use for this project, is a set of simple object-oriented classes with implementations in both Java and C++. You are free to use either of the implementations.

The classes include:

- `simpleCanvas` – a simple 2D canvas that allows the ability to set a pixel.
- `cglCanvas` – a subclass of `simpleCanvas` that includes methods that you will need to implement.
- `midtermMain` – the main function for the application.

It is expected that you have working versions of the functions `drawPolygon()` and `clipPolygon()` which were assigned in previous assignments. It is particularly important that you have a working `drawPolygon()` as you will not be able to generate any output without it.

If you do not have a working version of `drawPolygon()`, there is a method within the `simpleCanvas` class called `drawPoly()` that you can use for the project. Note that this routine will not fill in the polygon, rather only draw the outline. You should only use this routine if you do not have a working version of the `drawPolygon()` method within your `Rasterizer`.

All of the functions that you'll need to implement are found and documented in the `cglCanvas` class.

Note that you are free to create any additional auxiliary classes that you might need, however, you should not modify the code for `simpleCanvas` or `midtermMain`. You are encouraged to use the classes `Rasterizer` and `clipper` that you developed in your previous assignments, though this is not a requirement.

You can find the files in the content area of myCourses. Look under the resources category. Note that the distribution contains folders for both C++ and Java versions of the programming environment.

Additional Libraries:

You are welcome to use auxiliary libraries from the Web for matrix manipulation (or alternately, you can write your own). There are several good libraries available for both Java and C++. Here is a sample, though you are free to use whatever library you like (as long as it is freely available).

- Java: <http://math.nist.gov/javanumerics/jama/>
- C++: <http://www.techsoftpl.com/matrix/>

The Java implementation of the programming environment makes use of the standard `awt` and `Java2D` classes to access the 2D drawing areas, so no additional libraries or Java files, other than the ones provided are required.

However, since C++ does not have a standard 2D graphics environment, you will need some additional libraries if you choose to use the C++ implementation for your submission. The C++ implementation is built using SFML (Simple and Fast Multimedia Library), a free, cross-platform and lightweight library for performing common multimedia tasks. It can be downloaded at <http://www.sfml-dev.org/download.php>. Version 1.6 of the libraries are used. There are precompiled binaries for Windows, Mac, and Linux. NOTE: For Windows, there are precompiled binaries for Visual Studio 2005, and Visual Studio 2008. If you are using Visual Studio 2010, a precompiled version of the libraries is available in the content area of myCourses. Look under the resources category.

Please be sure to refer to the installation instructions for your given platform:

- Windows (Visual Studio) - <http://www.sfml-dev.org/tutorials/1.6/start-vc.php>
- Mac OS X -- <http://www.sfml-dev.org/tutorials/1.6/start-osx.php>
- Linux -- <http://www.sfml-dev.org/tutorials/1.6/start-linux.php>

Submission:

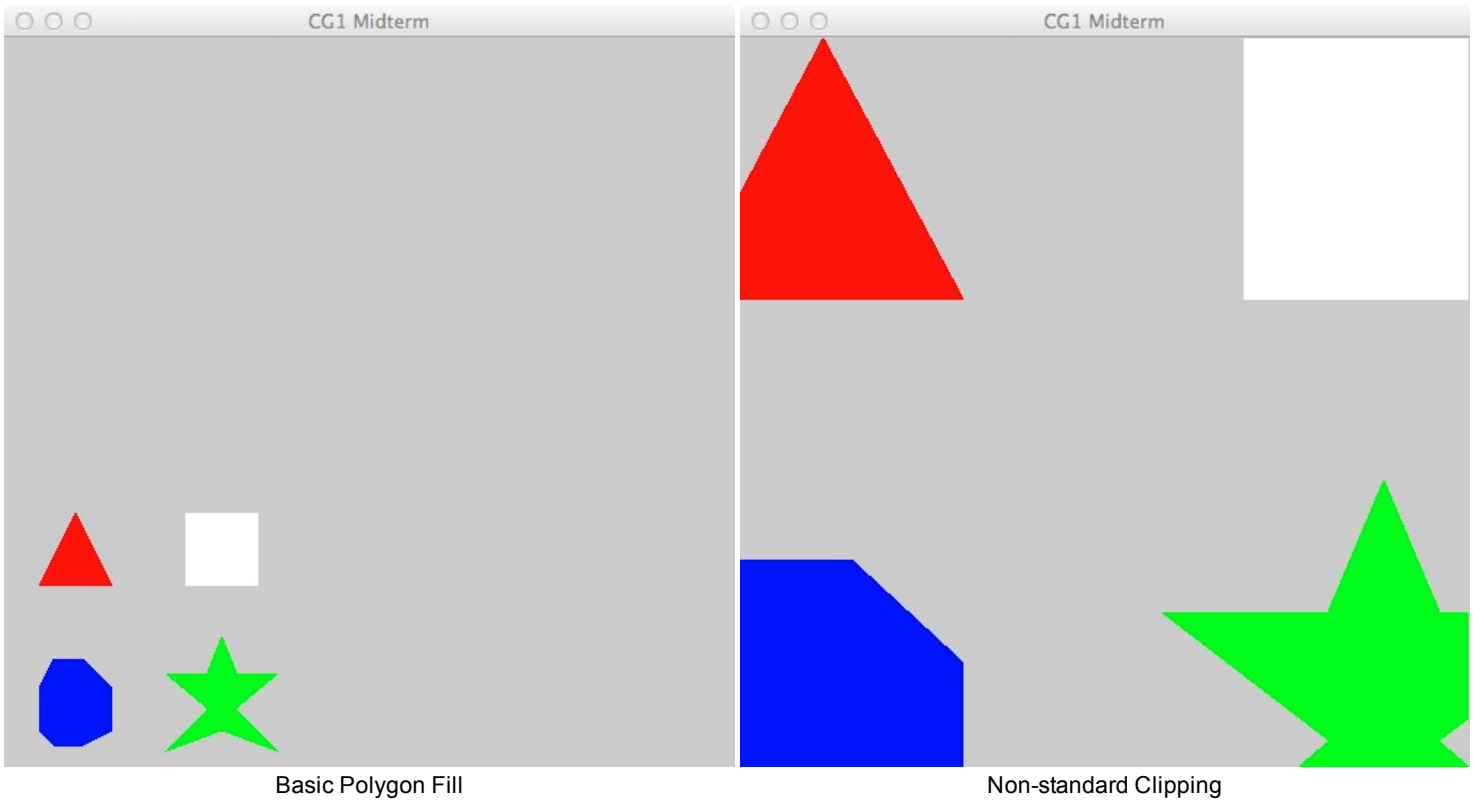
Your submission should only include the files for the `cg1Canvas` class that you have modified and any other auxiliary classes that you have created. For the Java version, this would mean the file `cg1Canvas.java` and for the C++ version, this means the files `cg1Canvas.cpp` and `cg1Canvas.h`. **If you used an external Matrix library, please include a README on where I can find the library, or include required files with your submission.** PLEASE DO NOT SUBMIT code for `simpleCanvas` or `midtermMain` as you should not be changing the code for these classes.

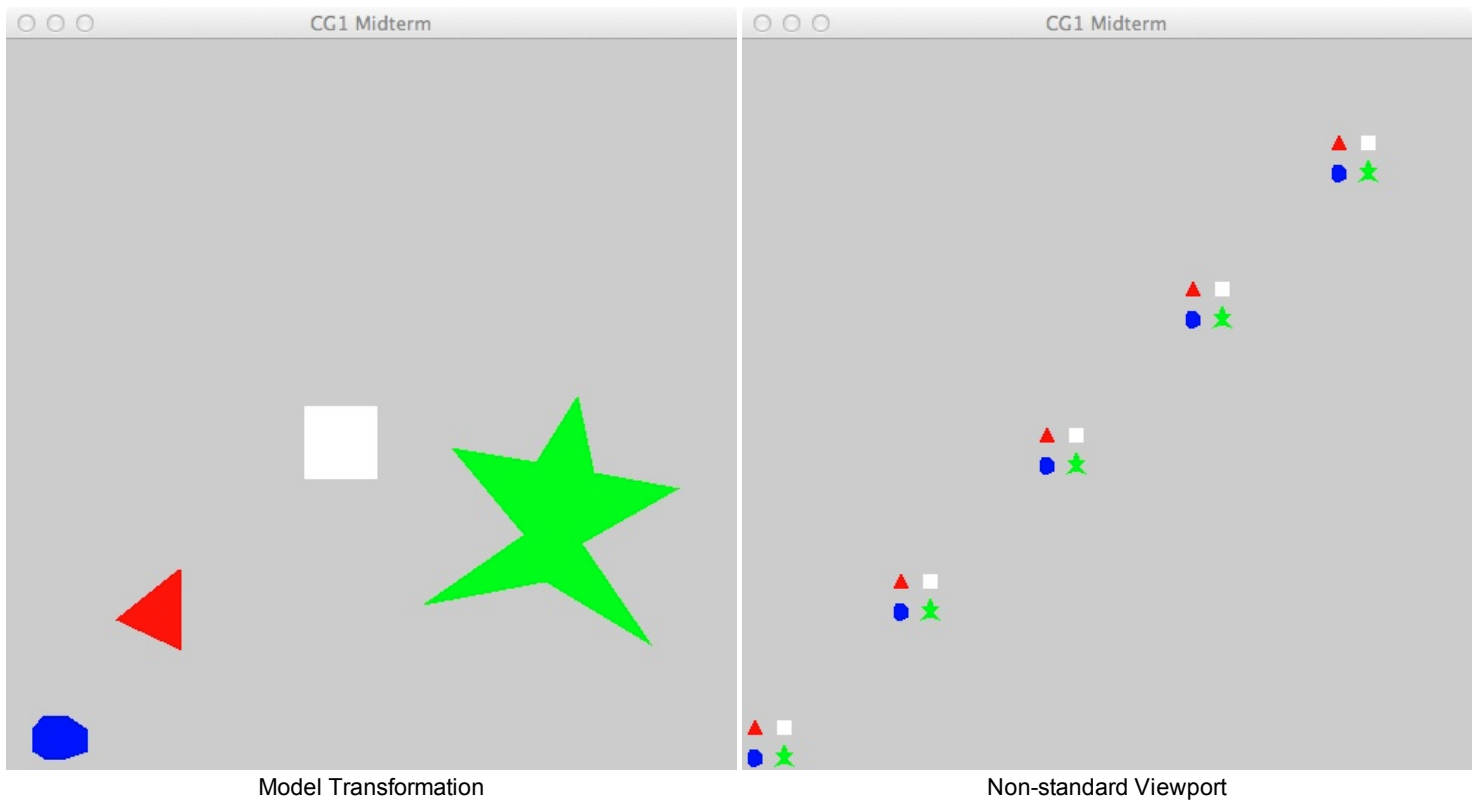
Submissions should be made to the dropbox labeled Midterm Project Option

Note that late submissions will not be accepted. There will also be no resubmits.

Output:

The test program consists of 4 screens (shown below). The first tests basic polygon drawing, the second tests use of a non-standard clip window, the 3rd tests model transformations and the last tests the use of non-standard viewports.





Model Transformation

Non-standard Viewport

Grading:

This midterm project option is worth 100 points distributed as follows:

- 30 points for submitting code that compiles, runs, and produces a window.
- 30 points for basic polygon drawing (10 points only if you use the supplied `drawPoly()` routine in `simpleCanvas`)
- 20 points for model transformations
- 10 points for non-standard clipping
- 10 points for non-standard viewport